Smart Environmental Safety Monitoring System

Benni Dorfman, T251292

I . INTRODUCTION

Background and Context

Work areas, storehouses and enclosed environments require real-time environmental monitoring to ensure safety of personnel and detect hazards early. Environmental factors such as toxic gas accumulation, extreme temperature spikes, humidity fluctuations, and poor illumination can endanger health and damage sensitive electronic equipment. Traditional manual checks are irregular and fail to provide the immediate response necessary during rapid hazard onset.

Aims and Objectives

The aim of this project was to develop a microcontroller-based Smart Environmental Safety Monitoring System (SESMS) using an STM32 microcontroller. The system continuously checks environmental parameters, displays and broadcasts environmental data and activates safety warnings when necessary.

The system must be able to:

- Continuously measure temperature, relative humidity, atmospheric pressure, toxic/combustible gas concentration, and ambient light level.
- Display real-time data and environmental safety status on an I2C-driven OLED display.
- Activate safety alarms when monitored data breach predefined threshold.
- Transmit data over bluetooth.

II SYSTEM ARCHITECTURE

Hardware Overview

The hardware layout uses a STM32 controller as a base, with peripherals connected as inputs and outputs.

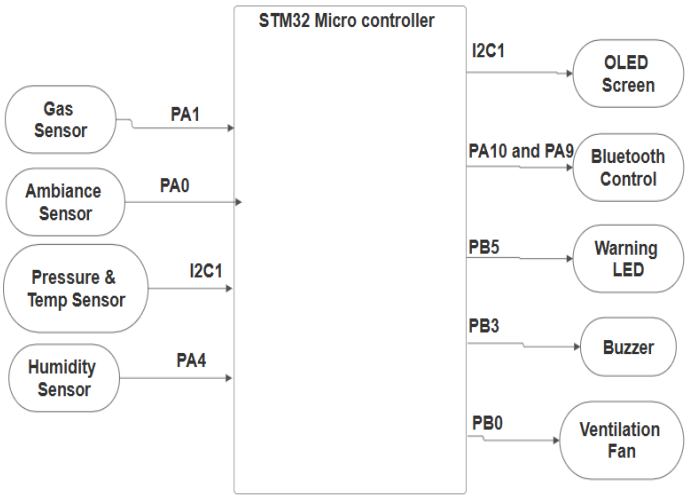


Figure 1: System Block Diagram

Component Allocation

Table 1: Components

Component	Hardware	Purpose
Micro controller	STM32F401RE	Processing and Peripheral Handling
Gas	MQ-2	Measures air via for contaminants
Ambiance	Temt6000	Monitors area illumance
Pressure & Temp	BMP280	Checks for unsafe pressure and heat spikes
Humidity	DHT11	Makes sure area has good humidity
OLED Screen	SSD1306 OLED	Displays data
Bluetooth Control	HC-05 Bluetooth Transceiver	Sends data to receiver
Warning LED	Red LED	Visually warns workers of unsafe environment

Active Buzzer	TDB05LFPN	Audibly warns workers of unsafe environment
Ventilation Fan	HKD Raspberry Pi DC Fan	Cools down area and distributes dangerous gasses

I II FIRMWARE IMPLEMENTATION

Code Structure And Sources

The project was constructed in C within STM32CubeIDE. AI was used for debugging.

Table 2: File Structure

File	Purpose	Source
AmbientLight.c/.h	Uses ADC and the onboard LDR to calculate ambient illumination.	Custom-written driver
Backdrop.c/.h	Stores bitmap graphics data for custom OLED screens	Bitmaps generated through Piskel.com and image2cpp.com
bluetooth.c/.h	Manages communication with Bluetooth device	Custom-written driver
DHT11.c/.h	Uses a single-wire bidirectional communication protocol through microsecond timing to read 40-bit relative humidity packets.	Open-source Github driver by medpy87587 customized for system
fonts.c/.h	Contains font data needed for rendering text onto the display	Open-source Adafruit_SSD1306 graphics library
GasSensor.c/.h	Configures ADC1, calibrates against a clean-air baseline, and converts the raw ADC value to gas concentration percentage.	Custom-written driver
main.c/.h	Entry point, starts core system clocks and peripherals and starts the main loop.	Auto-generated by STM32CubeMX
ssd1306.c/.h	Hardware driver library for the 128 x 64 OLED display.	Open-source Adafruit_SSD1306 graphics library
OLED.c/.h	Handles screen rendering	Custom-written driver
SystemControls.c/.h	Runts through all stages of program, polls sensors and safety thresholds	Custom-written driver

Safety Thresholds

The safety parameters and boundaries are defined below:

- Ambiance: Unsafe when below 3%
- Temperature: Unsafe when above 35C
- Humidity: Unsafe when above 80%
- Pressure: Unsafe when below 95kPa
- Gasses: Unsafe when above 30%

IV TESTING METHODOLOGY

Testing was performed to validate sensor calibration, threshold triggers, and communication reliability.

Table 3: Test Cases

Case	Input Condition	Observed Result
ADC Multi Channel Conversion	Exposed Gas sensor and LDR to variable inputs	Stable through the entire range.
Temperature Changes	Fan placed next to sensor	Temperature slowly lowers
Low-Light Warning	LDR covered to simulate darkened room	Ambiance goes below 5% and activates warning triggers
Display Changes	Watching the screen without taking action	Displays all pages sequentially without freezing
Bluetooth Connection	Paired HC-05 with phone	Clean ASCII telemetry stream at 9600 baud

Bluetooth Receive	Phone sends the string ‘Temperature’ to the STM32	OLED Screen changes to the Temperature Screen and sends out temperature data to the phone.

V DISCUSSION:

Goal Realization

The Smart Environmental Safety Monitoring System designed for the STM32 satisfied all project objectives. The thresholds were successfully set and monitored. In practice they activated with a little latency. Wireless reporting over bluetooth was reliable and allowed the user to manage the display.

Limitations & Future Work

The hardware prototype performed well, although its reliability can be improved through using a pcb board or soldered circuit instead of a breadboard. More sensors could be added in the future for better monitoring. Changing bluetooth to wire will allow for longer range monitoring.

VI CONCLUSION:

A standalone Smart Environmental Safety Monitoring System was designed, coded, and physically validated using the STM32F401RE microcontroller. The platform achieved real-time acquisition of ambient light, toxic gas concentration, temperature, relative humidity, and atmospheric pressure. The project proved itself as a reliable embedded safety system suitable for laboratory and light-industrial safety compliance.

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